

Francesco Massafra

Machine Learning Engineer

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Summary

M.Sc. Artificial Intelligence student seeking a Machine Learning (ML) Engineer internship (Summer 2027): large-scale ML systems, multi-GPU inference on High-Performance Computing (HPC) clusters (A100/H100), Large Language Model (LLM) evaluation, profiling at scale.

Education

University of Amsterdam

M.Sc. Artificial Intelligence

- GPA: 8.8/10. Coursework: Machine Learning 1 (Teaching Assistant), Deep Learning 1, Natural Language Processing (NLP).
- Extra semester planned for a Summer 2027 industry internship.

University of Pisa

B.Sc. Computer Science, 110/110 (top 5%), thesis: ML biomarkers for multiple sclerosis

Amsterdam, Netherlands

Sept 2025 - Jan 2028

Pisa, Italy

Sept 2021 - Nov 2024

Experience

University of Amsterdam

Teaching Assistant, Machine Learning 1 (graduate course)

- Lead weekly working sessions and hands-on exercises for 50 students in the graduate Machine Learning 1 course.
- Grade assignments and exams for 200 students against the course rubric, returning written feedback.
- Hold office hours, supporting students on PyTorch implementation and debugging.

Amsterdam, Netherlands

Sept 2026 - Present

MEDICA Project, University of Pisa

Research Assistant

- Developed the preprocessing and ML pipeline for MS-vs-control classification over 10 GEO datasets (1,038 microarray samples; 146,798 single cells from 32 patients); co-first author, [manuscript](#) under review.
- Trained per-dataset XGBoost models with Bayesian search on leakage-safe patient-grouped splits; best AUC 0.94 / F1 0.90.
- Rebuilt feature attribution with vectorized SHAP, 60% faster; SHAP selection surfaced candidate immune-checkpoint biomarkers.

Pisa, Italy

Dec 2024 - Dec 2025

Wylit

Backend Developer

- Engineered Node.js/PostgreSQL backend services across 15+ client deployments: REST APIs serving 10K+ daily requests.
- Cut p95 latency by 40% via PostgreSQL query tuning and Redis caching.
- Implemented CI/CD test and container-build checks with service health alerting.

Hybrid

Nov 2024 - Sept 2025

Research Projects

Reproducing & Extending “Towards Safer Pretraining”

2026

N. Caselli, **F. Massafra** (co-first), S. Lo Sardo, et al. · *Under review (TMLR)* [\[code\]](#)

- Reimplemented the three-pillar harmful-content filtering framework from scratch; benchmarked 13 open-weight LLMs served via Ollama: Llama 3.3 70B at F1 0.80 on TTP-Eval (matching GPT-4o).
- Classified 3M webpages (Common Crawl, C4, FineWeb) with HarmFormer in 33.25 GPU-h on one A100 vs 105.88 GPU-h for the LLM classifier on 30K pages: $\sim 100\times$ throughput at one-third the GPU cost.

Scalable Semantic-ID Recommender with Parallel Decoding (RPG)

2026

N. Caselli, **F. Massafra**, I. Pantelidis, M. Thabet, P. van Wijk, S. Wu (supervisor) · technical report [\[code\]](#)

- Reproduced RPG across 6 Amazon domains ($\sim 2M$ interactions, up to 64K items): text-embedding-3-large vectors discretised with FAISS/OPQ into 16-64-token semantic IDs, decoded in parallel by independent per-digit transformer heads.
- Graph-constrained decoding holds 33.6s/epoch and 0.104 GiB peak CUDA memory as candidates grow 20K $\rightarrow$ 500K, where both SASRec baselines exceed 110s/epoch.

Hi-LeWM: Hierarchical Planning in LeWorldModel

2026

N. Caselli, **F. Massafra** (co-first), S. Punzo, S. Lo Sardo, I. Pantelidis, et al. · *World Models Workshop, 2026* [\[code\]](#)

- Built Hi-LeWM (30.5M parameters, 12.5M trainable): frozen LeWorldModel plus a transformer macro-action encoder and adaLN predictor, planned by two-scale CEM.
- Showed naive hierarchy hurts long-horizon control (-14 pp at $d=50$): unconstrained CEM picks off-support macro-actions; support-constrained CEM fixes it without retraining: 64.0% at $d=50$ ($+11.3$ pp) and 32.7% at $d=75$ ($+14.7$ pp).

Probing Video Foundation Models for Intuitive Physics

2026

N. Caselli, **F. Massafra** (co-first), S. Lo Sardo, et al. · *Under review, 2026* [\[code\]](#)

- Probed 13 frozen video backbones (V-JEPA, VideoMAE, LTX-Video; 86M-1.85B) with linear/MLP/temporal probes at 4 depths.
- Temporal-attentive probes recover the most signal (V-JEPA 95.0% pair consistency on MVP, VideoMAE 73.9% violation-of-expectation on IntPhys2) at intermediate-to-late depth; frame shuffling costs 25.3-48.3 pp.

Technical Skills

Languages: Python, C, C++, SQL, TypeScript, Bash

ML & Deep Learning: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, SHAP, NumPy/pandas, W&B

LLMs & Retrieval: local LLM serving & evaluation (Ollama), LangChain, FAISS/OPQ vector search, constrained decoding

Systems & HPC: Linux, Slurm, CUDA, multi-GPU training/inference (A100, H100), Docker, Git, CI/CD

Data & Backend: PostgreSQL, Redis, REST APIs, Node.js

Practice: pytest, profiling, benchmark design, Weights & Biases (W&B)